

XeLaTeX is a replacement for pdfLaTeX, developed to overcome certain problems associated with pdfLaTeX. Even though such solutions are available, we will be discussing two easy methods to process LaTeX files containing PSTricks code, one involving the Linux terminal and the other involving Texmaker, a LaTeX editor. In both these techniques, we directly produce output files in a format other than PDF (Portable Document Format) and then use some simple tools to convert those files into PDF format. Whether you use the terminal or Texmaker, the formatted text in the output document will be the same. As a side note, XeLaTeX is available in Texmaker.

A simple PSTricks example

Before moving on to our first PSTricks example, consider the simple LaTeX template shown below, which was introduced in the previous article that discussed PGF/TikZ.

```
\documentclass{article}
\begin{document}
  % Here you can embed your PSTricks code
\end{document}
```

Just like PGF/TikZ code, all our PSTricks code will also be embedded inside this minimal LaTeX code for further processing. The PSTricks code inside the document environment of LaTeX is marked by the lines of code ‘\begin{document}’ and ‘\end{document}’. In the given LaTeX template, we only have a comment shown by the following line of code:

```
% Here you can embed your PSTricks code
```

...inside the document environment. Please note that LaTeX comments begin with a percentage symbol and will not be displayed in the final output document.

Sometimes, it is better to study a new tool by comparing its functionalities with a tool with which we are familiar. The examples provided in this article are somewhat similar to the ones we have discussed in the previous article about PGF/TikZ. But remember, even if you have missed our discussion about PGF/TikZ, it will in no way affect your learning of PSTricks. Now consider the simple LaTeX file *pstricks1.tex* shown below containing PSTricks code, which draws two rectangles—just like our PGF/TikZ example. This and all the other scripts discussed in this article are available for download from opensourceforu.com/article_source_code/June18PSTricks.zip.

```
\documentclass{article}
\usepackage{pstricks}
\begin{document}
  \begin{pspicture}(15,15)
    \psline[linecolor=blue,linewidth=1pt](1,1)(5,1)(5,7)
    (1,7)(1,1)
```

```
\pspolygon[linecolor=red,linewidth=8pt](6,1)(10,1)
(10,7)(6,7)
\end{pspicture}
\end{document}
```

The Cartesian coordinate system of PSTricks is similar to that of PGF/TikZ. The origin denoted by (0,0) can be considered to be located at the bottom-left corner of the plane when displayed on the screen, which is similar to PGF/TikZ. The distance of any component can be measured in the number of pixels called points or in SI units of distance, say a metre or centimetre.

Now let us try to understand the working of the file *pstricks1.tex*. The second line of code ‘\usepackage{pstricks}’ makes sure that the functionalities of PSTricks can be used in this LaTeX script. As mentioned earlier, all the PSTricks code in a LaTeX script should be embedded inside the document environment of LaTeX marked by the commands ‘\begin{document}’ and ‘\end{document}’. All the PSTricks code to draw specific images in your LaTeX document should be embedded inside the *pspicture* environment marked by the commands ‘\begin{pspicture}(15,15)’ and ‘\end{pspicture}’.

The code fragment (15,15) denotes the top-right corner of the rectangle in which the PSTricks image will be drawn. The bottom-left corner can also be specified but, in this case, the default value of (0,0) is taken as this parameter. It is possible to have any number of *pspicture* environments inside the document environment of LaTeX — meaning a LaTeX document can contain any number of PSTricks images. The following line of code:

```
\psline[linecolor=blue,linewidth=1pt](1,1)(5,1)(5,7)(1,7)
(1,1)
```

...draws five lines. The first line is drawn from point (1,1) to point (5,1), followed by a line from point (5,1) to point (5,7), followed by a line from point (5,7) to point (1,7) and, finally, a line from point (1,7) to the starting point (1,1) so that a rectangle is drawn. The code fragment:

```
[linecolor=blue,linewidth=1pt]
```

...sets the colour of the line to blue, and sets the line width to one point.

The line of code:

```
\pspolygon[linecolor=red,linewidth=8pt](6,1)(10,1)(10,7)
(6,7)
```

...draws a polygon with five sides. The first line is drawn from point (6,1) to point (10,1), followed by a line from point (10,1) to point (10,7), followed by a line from point (10,7) to point (6,7) and since a polygon is closed, finally a line is drawn from point (6,7) to point (6,1) to complete the